

INHA UNIVERSITY TASHKENT

SPRING SEMESTER 2024

SOC 2040

SYSTEM PROGRAMMING

CREDITS/HOURS PER WEEK : 3/3

COURSE TYPE : TECHNICAL CORE SEQUENCE

INSTRUCTOR

DR. A. R. NASEER

HEAD & PROFESSOR OF COMPUTER SCIENCE & ENGG

INHA UNIVERSITY TASHKENT

SPRING SEMESTER 2024

SOC 2040

SYSTEM PROGRAMMING

CLASS SCHEDULE

GROUP1: ICE 22-01, 02, 03 MON 11.00 – 12.30 R. No. A202
TUE 09.30 – 11.00 R. No. A202

GROUP2: ICE 22-04, CSE 01,02,03 TUE 11.00 – 12.30 R. A202
THU 11.00 – 12.30 R. A202

GROUP3: CSE 04, 05, 06 WED 09.30 – 11.00 R. No. A202
THU 09.30 – 11.00 R. No. A202

GROUP4: CSE 07, 08, 09 MON 09.30 – 11.00 R. No. A202
WED 11.00 – 12.30 R. No. A202

OFFLINE CLASS RULES

TO BE FOLLOWED VERY STRICTLY

- 1. BE PRESENT IN THE CLASS ATLEAST 5 MINUTES BEFORE THE START OF THE CLASSES**
- 2. STUDENTS LATE BY MORE THAN 5 MINUTES WILL NOT BE ALLOWED TO ENTER THE CLASS**
- 3. USE OF CELLPHONES & LAPTOPS STRICTLY PROHIBITED INSIDE THE CLASS**
- 4. CELL PHONES & LAPTOPS SHOULD NOT BE KEPT ON THE DESK**
- 5. CELLPHONES SHOULD BE KEPT EITHER IN YOUR POCKET OR BACKPACK/BAG YOU CARRY**
- 6. SLEEPING IN CLASS IS TOTALLY PROHIBITED**
- 7. BRING A NOTE BOOK FOR THIS COURSE AND MAKE NOTES IN THIS BOOK DURING THE CLASS**
- 8. NO DRINKING & NO EATING IS ALLOWED IN CLASS (PLEASE NOTE THAT CLASSROOM IS NOT CANTEEN)**

SOC 2040 - SYSTEMS PROGRAMMING

TEAM FORMATION INSTRUCTIONS FOR LAB ASSIGNMENTS

- **ALL LAB ASSIGNMENTS ARE TO BE COMPLETED IN GROUPS OF 4 TO 6 STUDENTS.**
- **YOU ARE REQUIRED TO FORM TEAMS COMPRISING A TEAM LEADER AND MEMBERS FOR CARRYING OUT LAB ASSIGNMENT1.**
- **A TEAM CAN HAVE A MAXIMUM OF 6 MEMBERS.**
- **A TEAM CAN HAVE MEMBERS DRAWN FROM OTHER SECTIONS ALSO.**
- **TEAM LEADER HAS TO PROVIDE THE TEAM DETAILS USING THE TEMPLATE PROVIDED AT THE ECLASS.**
- **DULY-FILLED IN TEAM DETAILS MUST BE SUBMITTED BY EACH TEAM LEADER BY EMAIL TO MY OFFICIAL ACCOUNT n.abdulrahim@inha.uz BY MONDAY, 19TH FEBRUARY 2024.**
- **LATE SUBMISSIONS ARE NOT ENTERTAINED**
- **ADHERE TO THE DEADLINE STRICTLY**

INHA UNIVERSITY TASHKENT
SPRING SEMESTER 2024
SOC 2040
SYSTEM PROGRAMMING

COURSE
SYLLABUS

SOC 2040 SYSTEM PROGRAMMING

COURSE DESCRIPTION

SOC 2040 SYSTEM PROGRAMMING COURSE DESCRIPTION

This course is intended to provide the basic knowledge required to implement efficient system software in modern computer systems.

This course covers several technical topics, which include

**Data Representation,
Assembly Languages, Compilation, Linking,
Cache Memory, Memory Hierarchy, Virtual Memory,
I/O subsystems,
Process/Thread and Synchronization**

SOC 2040

SYSTEM PROGRAMMING

COURSE LEARNING OBJECTIVES & OUTCOMES

Course Objectives

1. Understand components that comprise Computer Systems and how they operate in detail

(Relation with Program Outcomes PO 1, 3)

2. Understand how to control Computer Systems from the perspective of Software Engineers

(Relation with Program Outcomes PO 4,5,7)

Program Outcomes (POs)

At the end of the course, students should have

PO1 :

An ability to apply knowledge of mathematics, basic science, engineering, and information technology to the solution of engineering problems

PO3 :

An ability to define and formulate engineering problems

PO4 :

An ability to apply latest information, research-based knowledge, and appropriate tools to the solution of engineering problems

PO5 :

An ability to design systems, components and processes within realistic constraints

PO7:

An ability to communicate effectively under diverse situations

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SYSTEM PROGRAMMING

PREREQUISITES :

[MANDATORY] C PROGRAMMING, DATA STRUCTURES

RECOMMENDED COURSES AFTER THIS COURSE:

OPERATING SYSTEMS, UNIX PROGRAMMING, EMBEDDED SYSTEMS

COURSE SOFTWARE OR TOOL

LINUX (UBUNTU LINUX DISTRIBUTION)



SOC 2040

SYSTEM PROGRAMMING

Text Book

➤ **Computer Systems : A Programmer's Perspective**

- **Randal E. Bryant and David R. O'Hallaron**
3rd Edition, Global Edition Prentice Hall 2016
ISBN 10:1-292-10176-8

SOC 2040 SYSTEM PROGRAMMING

LECTURE TYPE : LECTURE AND LAB PRACTICE

- This course is a Lab-oriented Class.
- The instructor gives 2 sessions one and half hours of lectures about technical topics related to Computer Systems and Systems Programming.
- Teaching materials include a detailed description of the design rationale of modern computer systems, along with practical codes that show how to write System Software to control Computer Systems.
- Based on this knowledge, students are asked to implement practical system software in labs
- Each lab Exercise is composed of set of questions with different points according to their difficulties.
- In order to get points assigned to individual questions, students should solve them and submit before the deadline of the Lab assignment submission.

Grading Policy

Methods of Evaluation :

Assessment will be made on the basis of one-Midterm Exam, One Final Examination, Attendance and also based on solving questions in lab assignments

Evaluation Criteria :

-
- | | | |
|-----------------------------|---|-----|
| ▪ Attendance(5%) + Quiz(5%) | : | 10% |
| ▪ Lab (Assignments) | : | 40% |
| ▪ Mid-Term Exam | : | 25% |
| ▪ Final Exam | : | 25% |
-

Total : 100%

- **GRADES AWARDED : A+, A0, B+, B0, C+, C0, D+, D0, F (FAIL)**

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Course Topics & Weekly Lecture Breakdown

I WEEK

Lecture Topic : Introduction (Chapter 1)

An overview of a Computer System and a brief description of major components that comprise the computer system, including compiler, linker, loader, cache, virtual memory and I/Os

Lab Objective : Brief Introduction to Lab environment for Systems Programming

- Understand Virtual Box and Install it on your machine
- Understand what the Unix/Linux Operating System is
- Install the Linux operating System (Ubuntu 16.04 or Higer) in VirtualBox
- Understand how to use Unix/Linux shells

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Course Topics & Weekly Lecture Breakdown

II WEEK

Lecture Topic : Data Representation (Chapter 2)

Concept of bits representation in computer systems and how to manipulate bits using C languages. It also covers how integers are represented in modern CPUs(x86) and conversion and casting on integers

Lab Objective : Brief Introduction to useful tools for Systems Programming

- Install essential packages for Systems Programming
 - gcc, gdb, ddd, vim/vi, ctags, so on
- How to use vim editors
- How to use gcc and how to compile your program
- How to use gdb/ddd to fix bugs in your program
- Introduce a concept of Make and Makefile

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Course Topics & Weekly Lecture Breakdown

III WEEK

Lecture Topic : Machine Language - Basic (Chapter 3 - sections 3.1 to 3.5)

History of Intel processors and their architectures, Brief introduction to C, Assembly and machine codes, A detailed description of assembly language for x86(32-bit & 64-bit) machine including Registers, operands, Move and Arithmetic instructions

Lab Objective : Implement and execute a simple C program

- Implement a simple Test C Program, compile it and run it on your system – printing “hello World”, addition/subtraction/multiplication/division of two numbers, calculating factorial numbers, Fibonacci sequence, etc
- Fix a bug using system tools like gdb – the test program some bugs and students should fix all the bugs using gdb

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Course Topics & Weekly Lecture Breakdown

IV WEEK

Lecture Topic : Machine Language – Advanced (Chapter 3 - sections 3.5 to 3.7)

Advanced topics related to assembly language in x86 machines which include condition codes, conditional branches/moves, loops, switch statements, and procedures, working of stack in assembly codes

Lab Objective : Implement programs that perform bit manipulation

- Converting decimal representation to binary presentation
- Printing LSB and MSB of given numbers
- Reverse bit orders of given numbers, and so on

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Course Topics & Weekly Lecture Breakdown

V WEEK

Lecture Topic : Memory Hierarchy (Chapter 6 - Sections 6.1 to 6.3)

Storage Technologies (SRAM, DRAM, Disk, Solid State Drives) and trends, Introduction to the concept of locality and caching in the memory hierarchy

Lab objective : Analyze machine codes, understand its behaviors, edit the machine codes to change the behaviors

- Analyze a given machine code and implement the same program using C language**
- Analyze a given machine code and find a password**

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Course Topics & Weekly Lecture Breakdown

VI WEEK

Lecture Topic : Cache Memory (Chapter 6 - Sections 6.4 to 6.6)

Architecture and operations of CPU cache memory (e.g., Direct mapped cache, Set associative cache), its impact on System performance, several system software-level optimization techniques that exploit the properties of CPU caches

Lab Objective : Improve a Matrix Multiplication application

- Optimize a matrix multiplication algorithm so that it maximally exploits CPU's cache organization.
- Points are assigned to students differently based on performance improvements

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Course Topics & Weekly Lecture Breakdown

VII WEEK

Lecture Topic : Linking

(Chapter 7 - Sections 7.1 to 7.11)

**Basic concept of linking, static and dynamic libraries,
Discussion on several linking techniques that are essential to
implement system software**

Lab Objective : Understand Static/Dynamic Libraries

- **Write a C program that uses static/dynamic libraries**
- **Write a C program that uses inter-positioning features at a link time to detect memory leak problems**

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Course Topics & Weekly Lecture Breakdown

VIII WEEK

MID-TERM EXAM



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Course Topics & Weekly Lecture Breakdown

IX WEEK

Lecture Topic : Exceptional Control Flow (Chapter 8 - Sections 8.1 to 8.4)

Concept of Exceptions, including Traps, faults and aborts and why they are required. Then introducing the creation and termination of UNIX processes using UNIX system calls like `fork()`, `exit()`, `wait()`, and so on

Lab Objective : Implement a simple shell program that accepts user's command and run program

- Implement a simple shell program that accepts user commands from the command line
- Enhance the shell that executes a process based on the user's command

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Course Topics & Weekly Lecture Breakdown

X WEEK

Lecture Topic : Exceptional Control Flow – Signals (Chapter 8 – Sections 8.5 to 8.6)

Details of UNIX signals and nonlocal jumps that are core methods for software level exception handling

Lab Objective : Enhance the simple shell program

- Enhance the shell so that it supports background jobs
- Enhance the shell to support a kill command
- Enhance the shell to support a fg command

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Course Topics & Weekly Lecture Breakdown

XI WEEK

Lecture Topic : Virtual Memory – Concept (Chapter 9 - Sections 9.1 to 9.6)

Concept of virtual/physical address space and a virtual memory system, Important aspects of virtual memory systems as a tool, (e.g., a tool for memory management, a tool for caching, and a tool for memory protection). A detailed discussion on how address translation mechanisms work

Lab Objective : Understand how signals and nonlocal jumps work

- Implement a simple program that handles signals
- Implement a try-catch semantic in C programs using nonlocal jumps mechanism

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Course Topics & Weekly Lecture Breakdown

XII WEEK

**Lecture Topic : Virtual Memory – Memory Allocation
(Chapter 9 - Sections 9.8 to 9.9)**

Memory Allocation in Computer Systems and introducing a simple memory allocation policy such as implicit free lists

Lab Objective : Implement a set of memory management functions

- **Implement a smart pointer APIs that support the detection of memory leak, the detection of illegal memory access beyond allocated memory space, the detection of illegal free operations**
- **Combine it into the try-catch semantic implemented in the previous lab assignment**

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Course Topics & Weekly Lecture Breakdown

XIII WEEK

Lecture Topic : System-Level I/O

(Chapter 10 – Sections 10.1to 10.8)

Several topics related System-level I/O, including UNIX I/O, unbuffered/buffered I/O, sharing/redirection, and standard I/O

Lab Objective : Implement programs handling files for different purposes

- Implement a file copy program
- Implement a file encoding and decoding program
- Implement a file merging/splitting program

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Course Topics & Weekly Lecture Breakdown

XIV WEEK

Lecture Topic : Thread and Synchronization (Chapter 12 - Sections 12.1 to 12.6)

Concept of a Thread and data sharing between processes and threads. It covers synchronization techniques for mutual exclusion after introducing a race condition problem

Lab Objective : Understand Threads and how to use pthread APIs

- Write a simple application that uses pthread APIs
- Convert a serial problem to a parallel program using pthread APIs and then evaluate the performance improvement.
- Enhance the program above using semaphore so that a race condition can be avoided

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Course Topics & Weekly Lecture Breakdown

XV WEEK

FINAL EXAM



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Course Topics & Weekly Lecture Breakdown

XVI WEEK

MAKEUP ASSESSMENT



Contact Details :

E-mail : n.abdulrahim@inha.uz
dr_arnaseer@hotmail.com

Room No. : 402/4

Office Hours :

Wednesday - 2.00pm to 4.00pm
Thursday - 2.00pm to 4.00pm
or by appointment

Attendance Policy

- **All Regular students should attend all classes and laboratory sessions.**
- **Students who are absent without valid reason for more than 25% of the assigned hours for the course will be given a Grade “F”**

Academic Dishonesty Policy

- In order for instructors to fairly assess the quality and quantity of a student's learning (through course grades) as determined by work that students represent as their own, a relationship of trust between instructor and student is essential.
- Because violations of academic integrity most often involve, but are not limited to, efforts to deceive instructors, they represent a breach of the trust relationship between instructor and student, and undermine the core values of the university.
- For these reasons, the University and its instructors treat issues of academic dishonesty as serious violations of academic trust, and conduct rigorous investigations of students suspected of committing such acts.

ACTS OF ACADEMIC DISHONESTY INCLUDE, BUT ARE NOT LIMITED TO, THE FOLLOWING:

- the illegitimate use of materials in any form during a quiz or examination
- copying answers from the quiz or examination paper of another student
- plagiarizing (submitting as one's own ideas the work of another) or falsifying materials or information used in the completion of any assignment which is graded or evaluated as the student's individual effort
- submitting the same work for more than one course without the consent of the instructors of each course in which the work is submitted
- copying material from a web page and submitting it as one's own work
- quoting extensively from a document without making proper references to the source



If a student is found committing such acts in a quiz or home assignment or exam or term paper or term project he/she will be given a grade 0 in that part of the course.

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WISH YOU ALL

THE BEST

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